FRE-GT-9743 Assignment 4
Question 1

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Question 1: Why OTC prices carry a discount factor but futures prices do not

Key idea

The daily cash flows look identical, but they do different jobs. In a futures contract the daily payment *is* the profit or loss: it is paid out for good and nothing is left in the contract. In a collateralised OTC trade the daily payment is only a *deposit* that secures the trade: the profit or loss stays locked inside the contract until maturity T. Discounting is needed exactly for value that is still waiting to be paid in the future, so futures need one day of discounting (which cancels) and OTC needs discounting all the way to T.

Assumptions

Settlement dates $t_0 < t_1 < \cdots < t_n = T$. The one-period discount factor $D(t_k, t_{k+1})$ is known at t_k (rates are set at the start of each period). OTC trades are fully collateralised, $C(t_k) = V(t_k)$, and collateral earns interest at the same rate used in D.

Step 1: One equation for both contracts

Over one period, the value of any position today equals the discounted expectation of *(cash received at t_{k+1}) + (value still left in the contract at t_{k+1})*:

$$V(t_k) \;=\; D(t_k, t_{k+1})\, \mathbb{E}^{\mathbb{Q}}[\, X_{k+1} + V(t_{k+1}) \mid \mathcal{F}_{t_k}]\,. \tag{$*$}$$

The two contracts differ only in what X_{k+1} and $V(t_{k+1})$ are.

Step 2: Collateral adds no value

Posting collateral C at t_k and receiving it back with interest, $C/D(t_k, t_{k+1})$, at t_{k+1} has present value

$$D(t_k, t_{k+1}) \cdot \frac{C}{D(t_k, t_{k+1})} \;-\; C \;=\; 0.$$

So the OTC collateral flows are a fair loan and drop out of $(*)$. They remove credit risk but do not change what the contract is worth.

Step 3: Futures (the payment is final)

- Cash received: $X_{k+1} = F(t_{k+1}, T) - F(t_k, T)$.
- Value left after settlement: $V(t_{k+1}) = 0$ (the contract resets).
- Value today: $V(t_k) = 0$ (entering is free).

Substituting into $(*)$:

$$0 = D(t_k, t_{k+1})\Big(\mathbb{E}^{\mathbb{Q}}\big[F(t_{k+1}, T) \mid \mathcal{F}_{t_k}\big] - F(t_k, T)\Big).$$

$D(t_k, t_{k+1})$ is known and strictly positive, so it can be divided out:

$$F(t_k, T) = \mathbb{E}^{\mathbb{Q}}\big[F(t_{k+1}, T) \mid \mathcal{F}_{t_k}\big] \implies F(t, T) = \mathbb{E}^{\mathbb{Q}}\big[S(T) \mid \mathcal{F}_t\big].$$

The discount factor cancels because it multiplies zero: nothing is left in the contract to discount.

Step 4: OTC forward (the payment is a deposit)

Take a forward paying $S(T) - K$ at T.

- Cash received from the contract before T: $X_{k+1} = 0$ (collateral flows have zero value by Step 2).
- Value left after the margin call: $V(t_{k+1}) \neq 0$ (the contract is still alive).

Substituting into $(*)$:

$$V(t_k) = D(t_k, t_{k+1})\, \mathbb{E}^{\mathbb{Q}}\big[V(t_{k+1}) \mid \mathcal{F}_{t_k}\big].$$

Now the right-hand side is not zero, so D does *not* cancel. Iterating with the tower property, and using $D(t_k, t_{k+1})D(t_{k+1}, T) = D(t_k, T)$ and $V(T) = S(T) - K$:

$$V(t) = \mathbb{E}^{\mathbb{Q}}\big[\, D(t, T)\,(S(T) - K) \;\big|\; \mathcal{F}_t\big].$$

Each period contributes one discount factor, and they compound into $D(t, T)$.

Intuition in one line

Futures pay out your profit as it happens, so there is never a future amount to discount. OTC collateral is like a security deposit that earns interest and comes back, while the real payoff still waits at T, so it must be discounted.

Consequence: forward vs. futures price

Setting $V(t) = 0$ gives the fair forward price $K^* = \mathbb{E}^{\mathbb{Q}}[D(t, T)S(T)] \,/\, \mathbb{E}^{\mathbb{Q}}[D(t, T)]$, so

$$K^* - F(t, T) = \frac{\mathrm{Cov}^{\mathbb{Q}}\big(D(t, T),\, S(T)\big)}{\mathbb{E}^{\mathbb{Q}}[D(t, T)]}.$$

With deterministic rates the two prices coincide. If S rises when rates rise, then $\mathrm{Cov}(D, S) < 0$ and $F > K^*$: futures gains are received early and reinvested at high rates, while losses are financed at low rates.